

Project Proposal - CSE256

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1 Topic Introduction and Background

This survey will primarily focus on the topic of machine translation (MT). Translation tools like Google Translate have been an integral part of our society for the past few decades. Early MT tools have primarily been utilized to translate words and eventually simple phrases, but were still relatively inaccurate with more complex linguistics and sentences [1]. However, in 2016, Google pushed out their most notable update to their language translation platform using non-traditional methods of translating. This innovative change would revolutionize future approaches of machine translation models to allow for translating more complex structures while achieving a relatively high accuracy to retain meaning.

MT is typically categorized into either rule-based or corpus-based methods. Rule-based approaches utilized manual human writing to create the basic translation between two languages and typically required a physical dictionary for reference. While this approach was tedious, it was the primary method for early translation tools due to its simplicity. The 1990s saw the initial proposal of statistical machine translation (SMT) models to allow for automatic translation; however, due to its complexity and reliance on large bilingual datasets it failed to gain any significant adaptations until 1999 with the development of the "Egypt" toolkit. [1]

Phrase-based SMT methods were introduced in 2003 with "Pharaoh" and "Moses" as the primary innovations. Three years later, Google released their own phrase-based SMT translation tool, Google Translate. These milestones have shown a great improvement in the development of MT methods over the years and sparked further research to improve its quality. [1] One drawback is that SMT is a basic statistical natural language model and is reliant on parallel corpora to facilitate translations. It uses n-grams to decide the probability of the input and output matching in meaning. Translations were produced based on tables containing the probabilities of pre-trained existing and unseen generated translations, ultimately making it susceptible to erroneous and nonsensical "literal" translations. Although systems were developed to weed out highly improbable translations, they were still observable in early translation models like Google Translate. [2] This was a common issue with methods like SMT until neural machine translation (NMT) models were proposed in 2014.

2 Survey Task

This survey is specifically exploring the task of the mentioned NMT methods, particularly Google's Neural Machine Translation (GNMT) system employed by the new version of Google Translate. NMT directly learns the input and associated output texts and typically utilize recurrent neural networks (RNN) [2]. Attention is also utilized to help deal with longer phrasing and sentences, which traditional SMT models struggled with [3]. Google's research has already pointed out certain logistical issues such as training time, difficulty training with rare words, and failure to fully translate certain input sentences [3]. This survey aims to analyze GNMT's model and assess its performance to find additional limitations and propose possible solutions to them.

3 Relevant Papers

The first paper is *Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation* by Google consisting of the details of their GNMT model [3]. This paper introduces the

background behind their non-traditional NMT approach to solve common MT issues and comparisons with traditional SMT models. It dives into the sequence-to-sequence learning framework of their long short-term memory (LSTM) RNNs along with statistical explanations of their architectural design. The paper explains the functionality of each component— encoder, connections, etc.— along with details on their training and validation process. This paper will be the primary paper in this survey in evaluating the GNMT model and will serve as one of the main references to analyze and understand their GNMT code.

The second paper is *Progress in Machine Translation* by Haifeng Wang et al. containing an overview of the development of MT models [1]. This paper will provide the historical progress of MT throughout the late 1900s to 2021. An overview is provided of early rule-based models and the development of more recent corpus-based models. The SMT model is briefly explored as a background to the revolutionary improvements NMT models introduced. The paper also provides a theoretical overview of the fundamentals of NMT and the importance of attention and RNNs in providing high accuracy and bilingual evaluation understudy (BLEU) scores. This paper will also help as a basis in considering possible limitations of the GNMT model not mentioned in Google’s paper.

The third paper is *Statistical machine translation in the translation curriculum: Overcoming obstacles and empowering translators* by Dorothy Kenny et al. discussing the mechanics of SMTs [2]. This paper focuses on explaining the fundamentals of basic SMTs and its probabilistic language model. It introduces key elements of SMT like their translation and phrase tables and n-grams and how traditional models built translations off them. The paper also dives into historically important automatic evaluation metrics (AEMs) such as BLEU as the primary method of evaluating MT model performances.

4 Code and Proposition

The code that will be used for this survey is provided by NVIDIA’s GNMT model ¹ derived from Google’s GNMT [4] . The main purpose of this codebase is to implement a more efficient version of Google’s GNMT architecture with updated Python modules and parameters. This is to achieve the task of an NMT based MT model to provide more accurate and reliable translation of human languages from source to target. The codebase utilizes the same architecture as Google’s with a shifted attention module to after the first LSTM layer. The rest of the implementation generally follows the described architecture and process mentioned in the paper including the importance of GPU parallelization for training. [4]

A default configuration is provided by the repository. Both the encoder and decoder use shared embeddings and uses 4 LSTM layers instead of the 8 layers suggested by the paper, but this can be scaled in modified configurations. A Bahdanau attention mechanism is used to match the referenced mechanism in the Google paper. Mixed precision is used to train the model and utilizes new modules of PyTorch (specifically APEX) to provide certain computational benefits, making it easier to train on a variety of GPU architectures. [4]

The dataset the codebase used for training is the WMT16 English-German dataset² and will also be used for this survey. The main evaluation metric will be BLEU to examine the "humanness" of the translated results of the GNMT model. The training and validation loss per epoch is also calculated and used to evaluate the performance. These same metrics will be used for this survey to evaluate the performance and possible limitations of this codebase and architecture.

5 References

References

- [1] Haifeng Wang, Hua Wu, Zhongjun He, Liang Huang, and Kenneth Ward Church. Progress in machine translation. *Engineering*, 18:143–153, 2022.
- [2] Dorothy Kenny and Stephen Doherty. Statistical machine translation in the translation curriculum: Overcoming obstacles and empowering translators. *The Interpreter and Translator Trainer*, 8:276–294, 07 2014.

¹<https://github.com/NVIDIA/DeepLearningExamples/tree/master/PyTorch/Translation/GNMT>

²<https://www.statmt.org/wmt16/translation-task.html>

- [3] Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V. Le, Mohammad Norouzi, Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey, Jeff Klingner, Apurva Shah, Melvin Johnson, Xiaobing Liu, Łukasz Kaiser, Stephan Gouws, Yoshikiyo Kato, Taku Kudo, Hideto Kazawa, Keith Stevens, George Kurian, Nishant Patil, Wei Wang, Cliff Young, Jason Smith, Jason Riesa, Alex Rudnick, Oriol Vinyals, Greg Corrado, Macduff Hughes, and Jeffrey Dean. Google’s neural machine translation system: Bridging the gap between human and machine translation, 2016.
- [4] Nvidia. Gnmt v2 for pytorch.